

Thermodynamics

Instructor

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Chapter 14 – Problems

**Problem numbers correspond to
5th Edition**

14–15 A tank contains 21 kg of dry air and 0.3 kg of water vapor at 30°C and 100 kPa total pressure. Determine (a) the specific humidity, (b) the relative humidity, and (c) the volume of the tank.

Assumptions The air and the water vapor are ideal gases.

Analysis (a) The specific humidity can be determined from its definition,

$$\omega = \frac{m_v}{m_a} = \frac{0.3 \text{ kg}}{21 \text{ kg}} = \mathbf{0.0143 \text{ kg H}_2\text{O/kg dry air}}$$

(b) The saturation pressure of water at 30°C is

$$P_g = P_{\text{sat}@30^\circ\text{C}} = 4.2469 \text{ kPa}$$

Then the relative humidity can be determined from

$$\phi = \frac{\omega P}{(0.622 + \omega)P_g} = \frac{(0.0143)(100 \text{ kPa})}{(0.622 + 0.0143)(4.2469 \text{ kPa})} = \mathbf{52.9\%}$$

(c) The volume of the tank can be determined from the ideal gas relation for the dry air,

$$P_v = \phi P_g = (0.529)(4.2469 \text{ kPa}) = 2.245 \text{ kPa}$$

$$P_a = P - P_v = 100 - 2.245 = 97.755 \text{ kPa}$$

$$V = \frac{m_a R_a T}{P_a} = \frac{(21 \text{ kg})(0.287 \text{ kJ/kg} \cdot \text{K})(303 \text{ K})}{97.755 \text{ kPa}} = \mathbf{18.7 \text{ m}^3}$$

21 kg dry air
0.3 kg H₂O vapor
30°C
100 kPa

14–16 Repeat Prob. 14–15 for a temperature of 24°C.

Analysis (a) The specific humidity can be determined from its definition,

$$\omega = \frac{m_v}{m_a} = \frac{0.3 \text{ kg}}{21 \text{ kg}} = \mathbf{0.0143 \text{ kg H}_2\text{O/kg dry air}}$$

(b) The saturation pressure of water at 24°C is

$$P_g = P_{\text{sat @24°C}} = 2.986 \text{ kPa}$$

Then the relative humidity can be determined from

$$\phi = \frac{\omega P}{(0.622 + \omega)P_g} = \frac{(0.0143)(100 \text{ kPa})}{(0.622 + 0.0143)2.986 \text{ kPa}} = \mathbf{75.2\%}$$

(c) The volume of the tank can be determined from the ideal gas relation for the dry air,

$$P_v = \phi P_g = (0.752)(2.986 \text{ kPa}) = 2.245 \text{ kPa}$$

$$P_a = P - P_v = 100 - 2.245 = 97.755 \text{ kPa}$$

$$\mathcal{V} = \frac{m_a R_a T}{P_a} = \frac{(21 \text{ kg})(0.287 \text{ kJ/kg} \cdot \text{K})(297 \text{ K})}{97.755 \text{ kPa}} = \mathbf{18.3 \text{ m}^3}$$

21 kg dry air
0.3 kg H₂O vapor
24°C
100 kPa

14-17 A room contains air at 20°C and 98 kPa at a relative humidity of 85 percent. Determine (a) the partial pressure of dry air, (b) the specific humidity of the air, and (c) the enthalpy per unit mass of dry air.

Analysis (a) The partial pressure of dry air can be determined from

$$P_v = \phi P_g = \phi P_{\text{sat}@20^\circ\text{C}} = (0.85)(2.3392 \text{ kPa}) = 1.988 \text{ kPa}$$

$$P_a = P - P_v = 98 - 1.988 = \mathbf{96.01 \text{ kPa}}$$

(b) The specific humidity of air is determined from

$$\omega = \frac{0.622 P_v}{P - P_v} = \frac{(0.622)(1.988 \text{ kPa})}{(98 - 1.988) \text{ kPa}} = \mathbf{0.0129 \text{ kg H}_2\text{O/kg dry air}}$$

(c) The enthalpy of air per unit mass of dry air is determined from

$$\begin{aligned} h &= h_a + \omega h_v \cong c_p T + \omega h_g \\ &= (1.005 \text{ kJ/kg} \cdot ^\circ\text{C})(20^\circ\text{C}) + (0.0129)(2537.4 \text{ kJ/kg}) \\ &= \mathbf{52.78 \text{ kJ/kg dry air}} \end{aligned}$$

AIR
20°C
98 kPa
85% RH

14-18 Repeat Prob. 14-17 for a pressure of 85 kPa.

14-20 Determine the masses of dry air and the water vapor contained in a 240-m^3 room at 98 kPa, 23°C , and 50 percent relative humidity. *Answers: 273 kg, 2.5 kg*

Dew-Point, Adiabatic Saturation, and Wet-Bulb Temperatures

14–28 After a long walk in the 8°C outdoors, a person wearing glasses enters a room at 25°C and 40 percent relative humidity. Determine whether the glasses will become fogged.

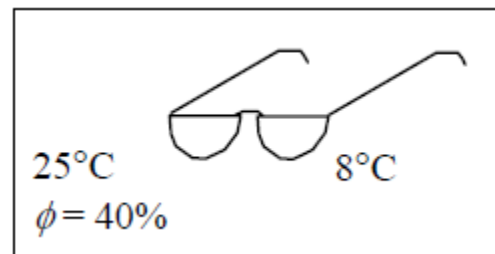
Analysis The vapor pressure P_v of the air in the house is uniform throughout, and its value can be determined from

$$P_v = \phi P_g @ 25^\circ\text{C} = (0.40)(3.1698 \text{ kPa}) = 1.268 \text{ kPa}$$

The dew-point temperature of the air in the house is

$$T_{\text{dp}} = T_{\text{sat}} @ P_v = T_{\text{sat}} @ 1.268 \text{ kPa} = \mathbf{10.5^\circ\text{C}} \quad (\text{from EES})$$

That is, the moisture in the house air will start condensing when the air temperature drops below 10.5°C. Since the glasses are at a lower temperature than the dew-point temperature, some moisture will condense on the glasses, and thus they **will get fogged**.



14–29 Repeat Prob. 14–28 for a relative humidity of 30 percent.

14-31 The dry- and wet-bulb temperatures of atmospheric air at 95 kPa are 25 and 17°C, respectively. Determine (a) the specific humidity, (b) the relative humidity, and (c) the enthalpy of the air, in kJ/kg dry air.

Analysis (a) We obtain the properties of water vapor from EES. The specific humidity ω_1 is determined from

$$\omega_1 = \frac{c_p(T_2 - T_1) + \omega_2 h_{fg2}}{h_{g1} - h_{f2}}$$

where T_2 is the wet-bulb temperature, and ω_2 is determined from

$$\omega_2 = \frac{0.622 P_{g2}}{P_2 - P_{g2}} = \frac{(0.622)(1.938 \text{ kPa})}{(95 - 1.938) \text{ kPa}} = 0.01295 \text{ kg H}_2\text{O/kg dry air}$$

95 kPa
25°C
 $T_{wb} = 17^\circ\text{C}$

Thus,

$$\omega_1 = \frac{(1.005 \text{ kJ/kg} \cdot ^\circ\text{C})(17 - 25)^\circ\text{C} + (0.01295)(2460.6 \text{ kJ/kg})}{(2546.5 - 71.36) \text{ kJ/kg}} = \mathbf{0.00963 \text{ kg H}_2\text{O/kg dry air}}$$

(b) The relative humidity ϕ_1 is determined from

$$\phi_1 = \frac{\omega_1 P_1}{(0.622 + \omega_1) P_{g1}} = \frac{(0.00963)(95 \text{ kPa})}{(0.622 + 0.00963)(3.1698 \text{ kPa})} = 0.457 \text{ or } \mathbf{45.7\%}$$

(c) The enthalpy of air per unit mass of dry air is determined from

$$\begin{aligned} h_1 &= h_{a1} + \omega_1 h_{v1} \cong c_p T_1 + \omega_1 h_{g1} = (1.005 \text{ kJ/kg} \cdot ^\circ\text{C})(25^\circ\text{C}) + (0.00963)(2546.5 \text{ kJ/kg}) \\ &= \mathbf{49.65 \text{ kJ/kg dry air}} \end{aligned}$$

14-32 The air in a room has a dry-bulb temperature of 22°C and a wet-bulb temperature of 16°C. Assuming a pressure of 100 kPa, determine (a) the specific humidity, (b) the relative humidity, and (c) the dew-point temperature. *Answers:* (a) 0.0090 kg H₂O/kg dry air, (b) 54.1 percent, (c) 12.3°C

14-35 Atmospheric air at 35°C flows steadily into an adiabatic saturation device and leaves as a saturated mixture at 25°C. Makeup water is supplied to the device at 25°C. Atmospheric pressure is 98 kPa. Determine the relative humidity and specific humidity of the air.

Analysis The exit state of the air is completely specified, and the total pressure is 98 kPa. The properties of the moist air at the exit state may be determined from EES to be

$$h_2 = 78.11 \text{ kJ/kg dry air}$$

$$\omega_2 = 0.02079 \text{ kg H}_2\text{O/kg dry air}$$

The enthalpy of makeup water is

$$h_{w2} = h_{f@25^\circ\text{C}} = 104.83 \text{ kJ/kg} \quad (\text{Table A - 4})$$

An energy balance on the control volume gives

$$h_1 + (\omega_2 - \omega_1)h_w = h_2$$

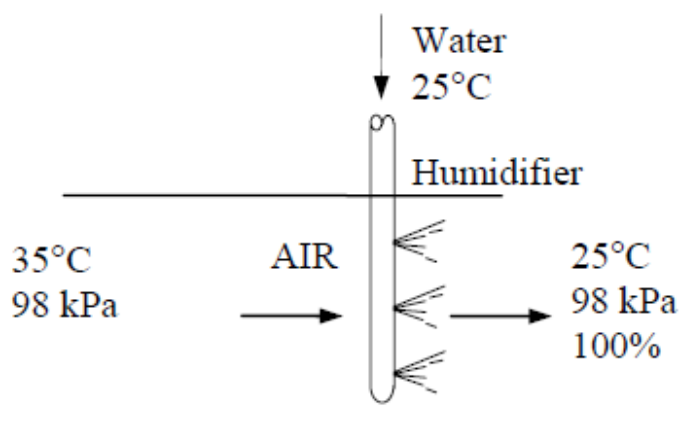
$$h_1 + (0.02079 - \omega_1)(104.83 \text{ kJ/kg}) = 78.11 \text{ kJ/kg}$$

Pressure and temperature are known for inlet air. Other properties may be determined from this equation using EES. A hand solution would require a trial-error approach. The results are

$$h_1 = 77.66 \text{ kJ/kg dry air}$$

$$\omega_1 = 0.01654 \text{ kg H}_2\text{O/kg dry air}$$

$$\phi_1 = 0.4511$$



14-40



The air in a room is at 1 atm, 32°C , and 60 percent relative humidity. Determine (*a*) the specific humidity, (*b*) the enthalpy (in kJ/kg dry air), (*c*) the wet-bulb temperature, (*d*) the dew-point temperature, and (*e*) the specific volume of the air (in m^3/kg dry air). Use the psychrometric chart or available software.

Human Comfort and Air-Conditioning

14-60 An average person produces 0.25 kg of moisture while taking a shower and 0.05 kg while bathing in a tub. Consider a family of four who each shower once a day in a bathroom that is not ventilated. Taking the heat of vaporization of water to be 2450 kJ/kg, determine the contribution of showers to the latent heat load of the air conditioner per day in summer.

14-61 An average (1.82 kg or 4.0 lbm) chicken has a basal metabolic rate of 5.47 W and an average metabolic rate of 10.2 W (3.78 W sensible and 6.42 W latent) during normal activity. If there are 100 chickens in a breeding room, determine the rate of total heat generation and the rate of moisture production in the room. Take the heat of vaporization of water to be 2430 kJ/kg.

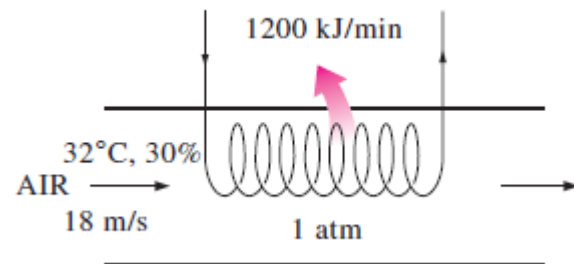
14-62 A department store expects to have 120 customers and 15 employees at peak times in summer. Determine the contribution of people to the total cooling load of the store.

14-64 For an infiltration rate of 1.2 air changes per hour (ACH), determine sensible, latent, and total infiltration heat load of a building at sea level, in kW, that is 20 m long, 13 m wide, and 3 m high when the outdoor air is at 32°C and 50 percent relative humidity. The building is maintained at 24°C and 50 percent relative humidity at all times.

14-65 Repeat Prob. 14-64 for an infiltration rate of 1.8 ACH.

Simple Heating and Cooling

14-70 Air enters a 40-cm-diameter cooling section at 1 atm, 32°C, and 30 percent relative humidity at 18 m/s. Heat is removed from the air at a rate of 1200 kJ/min. Determine (a) the exit temperature, (b) the exit relative humidity of the air, and (c) the exit velocity. *Answers: (a) 24.4°C, (b) 46.6 percent, (c) 17.6 m/s*



Analysis (a) The amount of moisture in the air remains constant ($\omega_1 = \omega_2$) as it flows through the cooling section since the process involves no humidification or dehumidification. The inlet state of the air is completely specified, and the total pressure is 1 atm. The properties of the air at the inlet state are determined from the psychrometric chart (Figure A-31) to be

$$h_1 = 55.0 \text{ kJ/kg dry air}$$

$$\omega_1 = 0.0089 \text{ kg H}_2\text{O/kg dry air} (= \omega_2)$$

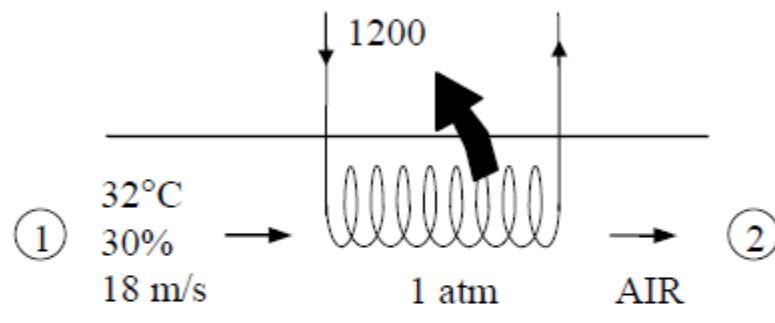
$$\nu_1 = 0.877 \text{ m}^3 / \text{kg dry air}$$

The mass flow rate of dry air through the cooling section is

$$\begin{aligned}\dot{m}_a &= \frac{1}{\nu_1} V_1 A_1 \\ &= \frac{1}{(0.877 \text{ m}^3 / \text{kg})} (18 \text{ m/s}) (\pi \times 0.4^2 / 4 \text{ m}^2) \\ &= 2.58 \text{ kg/s}\end{aligned}$$

From the energy balance on air in the cooling section,

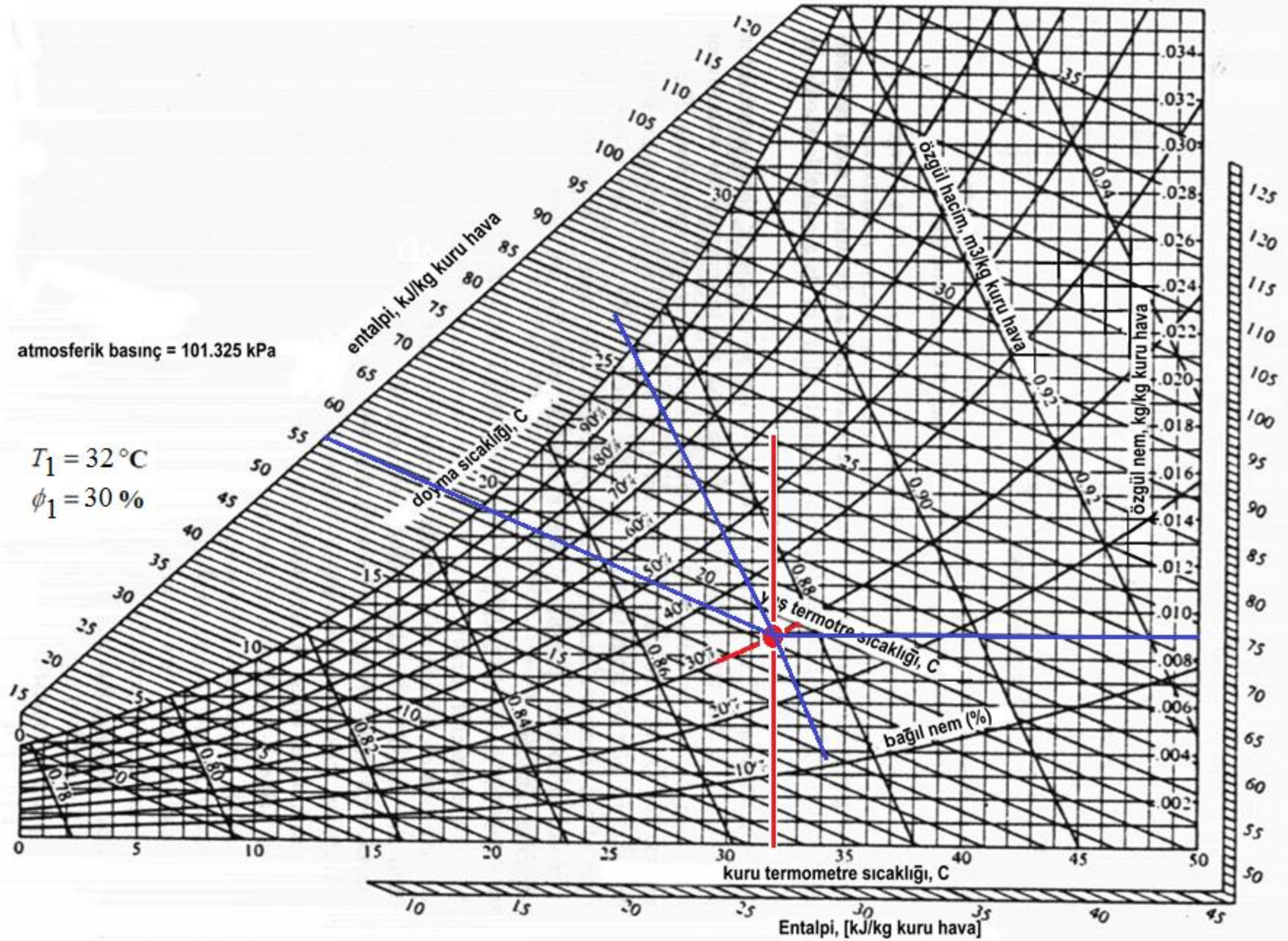
$$\begin{aligned}-\dot{Q}_{\text{out}} &= \dot{m}_a (h_2 - h_1) \\ -1200 / 60 \text{ kJ/s} &= (2.58 \text{ kg/s}) (h_2 - 55.0) \text{ kJ/kg} \\ h_2 &= 47.2 \text{ kJ/kg dry air}\end{aligned}$$



atmosferik basınç = 101.325 kPa

$T_1 = 32^\circ\text{C}$

$\phi_1 = 30\%$



The exit state of the air is fixed now since we know both h_2 and ω_2 . From the psychrometric chart at this state we read

$$T_2 = 24.4^\circ\text{C}$$

(b) $\phi_2 = 46.6\%$

$$\nu_2 = 0.856 \text{ m}^3 / \text{kg dry air}$$

(c) The exit velocity is determined from the conservation of mass of dry air,

$$\dot{m}_{a1} = \dot{m}_{a2} \longrightarrow \frac{\dot{V}_1}{\nu_1} = \frac{\dot{V}_2}{\nu_2} \longrightarrow \frac{V_1 A}{\nu_1} = \frac{V_2 A}{\nu_2}$$

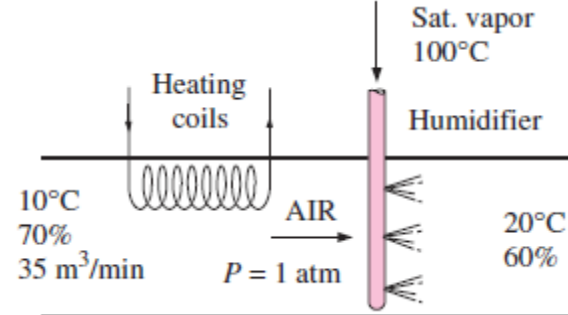
$$V_2 = \frac{\nu_2}{\nu_1} V_1 = \frac{0.856}{0.877} (18 \text{ m/s}) = 17.6 \text{ m/s}$$

14-71 Repeat Prob. 14-70 for a heat removal rate of 800 kJ/min.

Heating with Humidification

14-73 Air at 1 atm, 15°C, and 60 percent relative humidity is first heated to 20°C in a heating section and then humidified by introducing water vapor. The air leaves the humidifying section at 25°C and 65 percent relative humidity. Determine (a) the amount of steam added to the air, and (b) the amount of heat transfer to the air in the heating section. *Answers: (a) 0.0065 kg H₂O/kg dry air, (b) 5.1 kJ/kg dry air*

14-75 An air-conditioning system operates at a total pressure of 1 atm and consists of a heating section and a humidifier that supplies wet steam (saturated water vapor) at 100°C. Air enters the heating section at 10°C and 70 percent relative humidity at a rate of 35 m³/min, and it leaves the humidifying section at 20°C and 60 percent relative humidity. Determine (a) the temperature and relative humidity of air when it leaves the heating section, (b) the rate of heat transfer in the heating section, and (c) the rate at which water is added to the air in the humidifying section.



Assumptions 1 This is a steady-flow process and thus the mass flow rate of dry air remains constant during the entire process ($\dot{m}_{a1} = \dot{m}_{a2} = \dot{m}_a$). 2 Dry air and water vapor are ideal gases. 3 The kinetic and potential energy changes are negligible.

Properties The inlet and the exit states of the air are completely specified, and the total pressure is 1 atm. The properties of the air at various states are determined from the psychrometric chart (Figure A-31) to be

$$h_1 = 23.5 \text{ kJ/kg dry air}$$

$$\omega_1 = 0.0053 \text{ kg H}_2\text{O/kg dry air} (= \omega_2)$$

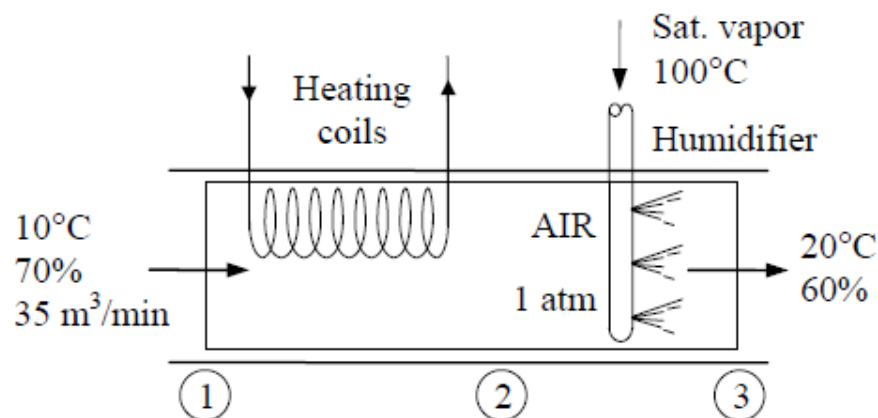
$$\nu_1 = 0.809 \text{ m}^3/\text{kg dry air}$$

$$h_3 = 42.3 \text{ kJ/kg dry air}$$

$$\omega_3 = 0.0087 \text{ kg H}_2\text{O/kg dry air}$$

Analysis (a) The amount of moisture in the air remains constant as it flows through the heating section ($\omega_1 = \omega_2$), but increases in the humidifying section ($\omega_3 > \omega_2$). The mass flow rate of dry air is

$$\dot{m}_a = \frac{\dot{V}_1}{\nu_1} = \frac{35 \text{ m}^3/\text{min}}{0.809 \text{ m}^3/\text{kg}} = 43.3 \text{ kg/min}$$



Noting that $Q = W = 0$, the energy balance on the humidifying section can be expressed as

$$\dot{E}_{\text{in}} - \dot{E}_{\text{out}} = \Delta \dot{E}_{\text{system}} \overset{\text{no (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\sum \dot{m}_i h_i = \sum \dot{m}_e h_e \quad \longrightarrow \quad \dot{m}_w h_w + \dot{m}_a h_2 = \dot{m}_a h_3$$

$$(\omega_3 - \omega_2) h_w + h_2 = h_3$$

Solving for h_2 ,

$$h_2 = h_3 - (\omega_3 - \omega_2) h_{g@100^\circ\text{C}} = 42.3 - (0.0087 - 0.0053)(2675.6) = 33.2 \text{ kJ/kg dry air}$$

Thus at the exit of the heating section we have $\omega_2 = 0.0053 \text{ kg H}_2\text{O dry air}$ and $h_2 = 33.2 \text{ kJ/kg dry air}$, which completely fixes the state. Then from the psychrometric chart we read

$$T_2 = 19.5^\circ\text{C}$$

$$\phi_2 = 37.8\%$$

(b) The rate of heat transfer to the air in the heating section is

$$\dot{Q}_{\text{in}} = \dot{m}_a (h_2 - h_1) = (43.3 \text{ kg/min})(33.2 - 23.5) \text{ kJ/kg} = \mathbf{420 \text{ kJ/min}}$$

(c) The amount of water added to the air in the humidifying section is determined from the conservation of mass equation of water in the humidifying section,

$$\dot{m}_w = \dot{m}_a (\omega_3 - \omega_2) = (43.3 \text{ kg/min})(0.0087 - 0.0053) = \mathbf{0.15 \text{ kg/min}}$$

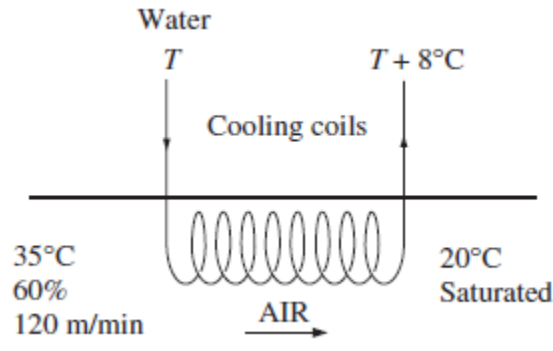
14-76 Repeat Prob. 14-75 for a total pressure of 95 kPa for the airstream. *Answers: (a) 19.5°C, 37.7 percent, (b) 391 kJ/min, (c) 0.147 kg/min*

Cooling with Dehumidification

14–79 An air-conditioning system is to take in air at 1 atm, 34°C , and 70 percent relative humidity and deliver it at 22°C and 50 percent relative humidity. The air flows first over the cooling coils, where it is cooled and dehumidified, and then over the resistance heating wires, where it is heated to the desired temperature. Assuming that the condensate is removed from the cooling section at 10°C , determine (a) the temperature of air before it enters the heating section, (b) the amount of heat removed in the cooling section, and (c) the amount of heat transferred in the heating section, both in kJ/kg dry air.



Air enters a 30-cm-diameter cooling section at 1 atm, 35°C, and 60 percent relative humidity at 120 m/min. The air is cooled by passing it over a cooling coil through which cold water flows. The water experiences a temperature rise of 8°C. The air leaves the cooling section saturated at 20°C. Determine (a) the rate of heat transfer, (b) the mass flow rate of the water, and (c) the exit velocity of the airstream.



Analysis (a) The saturation pressure of water at 35°C is 5.6291 kPa (Table A-4). Then the dew point temperature of the incoming air stream at 35°C becomes

$$T_{dp} = T_{sat} @ P_v = T_{sat} @ 0.6 \times 5.6291 \text{ kPa} = 26^\circ\text{C} \quad (\text{Table A-5})$$

since air is cooled to 20°C, which is below its dew point temperature, some of the moisture in the air will condense. The amount of moisture in the air decreases due to dehumidification ($\omega_2 < \omega_1$). The inlet and the exit states of the air are completely specified, and the total pressure is 1 atm. Then the properties of the air at both states are determined from the psychrometric chart (Fig. A-31) to be

$$h_1 = 90.3 \text{ kJ/kg dry air}$$

$$\omega_1 = 0.0215 \text{ kg H}_2\text{O/kg dry air}$$

$$\nu_1 = 0.904 \text{ m}^3/\text{kg dry air}$$

and

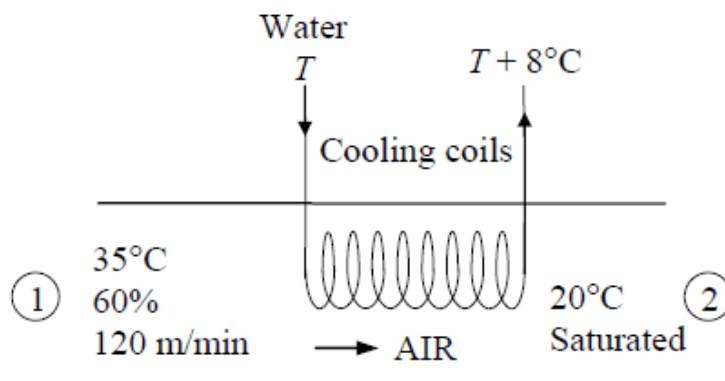
$$h_2 = 57.5 \text{ kJ/kg dry air}$$

$$\omega_2 = 0.0147 \text{ kg H}_2\text{O/kg dry air}$$

$$\nu_2 = 0.851 \text{ m}^3/\text{kg dry air}$$

Also, $h_w \cong h_f @ 20^\circ\text{C} = 83.93 \text{ kJ/kg}$ (Table A-4)

Then,



$$\dot{V}_1 = V_1 A_1 = V_1 \frac{\pi D^2}{4} = (120 \text{ m/min}) \left(\frac{\pi (0.3 \text{ m})^2}{4} \right) = 8.48 \text{ m}^3 / \text{min}$$

$$\dot{m}_{a1} = \frac{\dot{V}_1}{\nu_1} = \frac{8.48 \text{ m}^3 / \text{min}}{0.904 \text{ m}^3 / \text{kg dry air}} = 9.38 \text{ kg/min}$$

Applying the water mass balance and the energy balance equations to the combined cooling and dehumidification section (excluding the water),

$$\text{Water Mass Balance: } \sum \dot{m}_{w,i} = \sum \dot{m}_{w,e} \longrightarrow \dot{m}_{a1}\omega_1 = \dot{m}_{a2}\omega_2 + \dot{m}_w$$

$$\dot{m}_w = \dot{m}_a(\omega_1 - \omega_2) = (9.38 \text{ kg/min})(0.0215 - 0.0147) = 0.064 \text{ kg/min}$$

Energy Balance:

$$\dot{E}_{in} - \dot{E}_{out} = \Delta \dot{E}_{system} \overset{\text{no (steady)}}{=} 0 \longrightarrow \dot{E}_{in} = \dot{E}_{out}$$

$$\sum \dot{m}_i h_i = \sum \dot{m}_e h_e + \dot{Q}_{out} \longrightarrow \dot{Q}_{out} = \dot{m}_{a1}h_1 - (\dot{m}_{a2}h_2 + \dot{m}_w h_w) = \dot{m}_a(h_1 - h_2) - \dot{m}_w h_w$$

$$\dot{Q}_{out} = (9.38 \text{ kg/min})(90.3 - 57.5) \text{ kJ/kg} - (0.064 \text{ kg/min})(83.93 \text{ kJ/kg}) = \mathbf{302.3 \text{ kJ/min}}$$

(b) Noting that the heat lost by the air is gained by the cooling water, the mass flow rate of the cooling water is determined from

$$\dot{Q}_{cooling \text{ water}} = \dot{m}_{cooling \text{ water}} \Delta h = \dot{m}_{cooling \text{ water}} c_p \Delta T$$

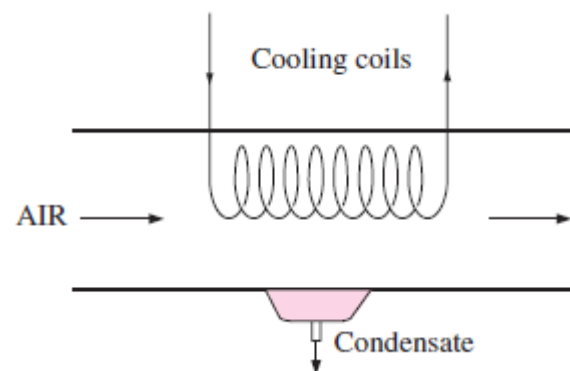
$$\dot{m}_{cooling \text{ water}} = \frac{\dot{Q}_w}{c_p \Delta T} = \frac{302.3 \text{ kJ/min}}{(4.18 \text{ kJ/kg} \cdot ^\circ\text{C})(8^\circ\text{C})} = \mathbf{9.04 \text{ kg/min}}$$

(c) The exit velocity is determined from the conservation of mass of dry air,

$$\dot{m}_{a1} = \dot{m}_{a2} \longrightarrow \frac{\dot{V}_1}{v_1} = \frac{\dot{V}_2}{v_2} \longrightarrow \frac{V_1 A}{v_1} = \frac{V_2 A}{v_2}$$

$$V_2 = \frac{v_2}{v_1} V_1 = \frac{0.851}{0.904} (120 \text{ m/min}) = \mathbf{113 \text{ m/min}}$$

14-86 Atmospheric air from the inside of an automobile enters the evaporator section of the air conditioner at 1 atm, 27°C and 50 percent relative humidity. The air returns to the automobile at 10°C and 90 percent relative humidity. The passenger compartment has a volume of 2 m³ and 5 air changes per minute are required to maintain the inside of the automobile at the desired comfort level. Sketch the psychrometric diagram for the atmospheric air flowing through the air conditioning process. Determine the dew point and wet bulb temperatures at the inlet to the evaporator section, in °C. Determine the required heat transfer rate from the atmospheric air to the evaporator fluid, in kW. Determine the rate of condensation of water vapor in the evaporator section, in kg/min.



14-87 Two thousand cubic meters per hour of atmospheric air at 28°C with a dew point temperature of 25°C flows into an air conditioner that uses chilled water as the cooling fluid. The atmospheric air is to be cooled to 18°C . Sketch the system hardware and the psychrometric diagram for the process. Determine the mass flow rate of the condensate water, if any, leaving the air conditioner, in kg/h. If the cooling water has a 10°C temperature rise while flowing through the air conditioner, determine the volume flow rate of chilled water supplied to the air conditioner heat exchanger, in m^3/min . The air conditioning process takes place at 100 kPa.

Analysis We may assume that the exit relative humidity is 100 percent since the exit temperature (18°C) is below the dew-point temperature of the inlet air (25°C). The properties of the air at the exit state may be determined from the psychrometric chart (Fig. A-31) or using EES psychrometric functions to be (we used EES)

$$h_2 = 51.34 \text{ kJ/kg dry air}$$

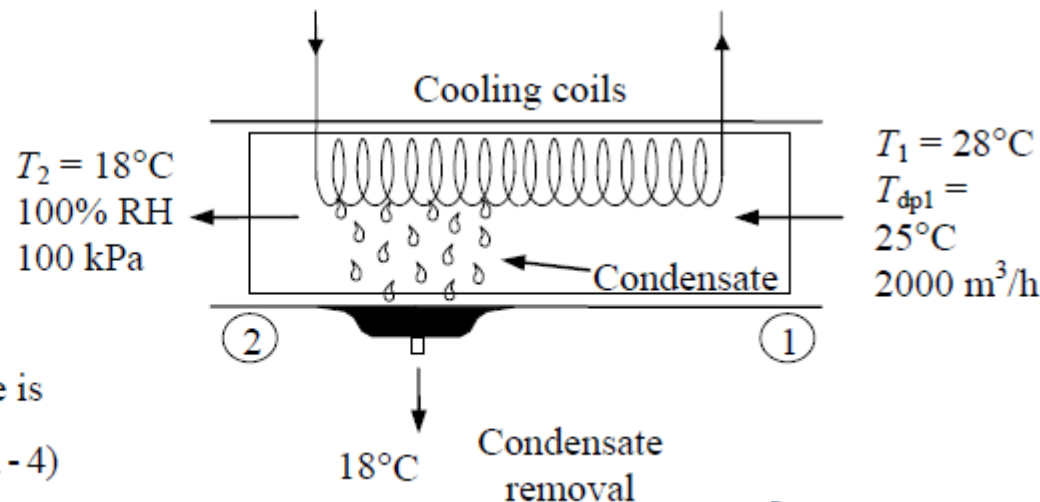
$$\omega_2 = 0.01311 \text{ kg H}_2\text{O/kg dry air}$$

The partial pressure of water vapor at the inlet state is (Table A-4)

$$P_{v1} = P_{\text{sat}@ } 25^{\circ}\text{C} = 3.17 \text{ kPa}$$

The saturation pressure at the inlet state is

$$P_{g1} = P_{\text{sat}@ } 28^{\circ}\text{C} = 3.783 \text{ kPa (Table A - 4)}$$



Then, the relative humidity at the inlet state becomes

$$\phi_1 = \frac{P_{v1}}{P_{g1}} = \frac{3.17}{3.783} = 0.8379$$

Now, the inlet state is also fixed. The properties are obtained from EES to be

$$h_1 = 80.14 \text{ kJ/kg dry air}$$

$$\omega_1 = 0.02036 \text{ kg H}_2\text{O/kg dry air}$$

$$\nu_1 = 0.8927 \text{ m}^3/\text{kg}$$

The mass flow rate of dry air is

$$\dot{m}_a = \frac{\dot{V}_1}{\nu_1} = \frac{(2000 / 60) \text{ m}^3/\text{h}}{0.8927 \text{ m}^3/\text{kg}} = 37.34 \text{ kg/min}$$

The mass flow rate of condensate water is

$$\begin{aligned}\dot{m}_w &= \dot{m}_a (\omega_1 - \omega_2) \\ &= (37.34 \text{ kg/min})(0.02036 - 0.01311) \\ &= 0.2707 \text{ kg/min} \\ &= \mathbf{16.24 \text{ kg/h}}\end{aligned}$$

The enthalpy of condensate water is

$$h_{w2} = h_{f@18^\circ\text{C}} = 75.54 \text{ kJ/kg} \quad (\text{Table A - 4})$$

An energy balance on the control volume gives



$$\dot{m}_a h_1 = \dot{Q}_{\text{out}} + \dot{m}_a h_2 + \dot{m}_w h_{w2}$$

$$(37.34 \text{ kg/min})(80.14 \text{ kJ/kg}) = \dot{Q}_{\text{out}} + (37.34 \text{ kg/min})(51.34 \text{ kJ/kg}) + (0.2707 \text{ kg/min})(75.54 \text{ kJ/kg})$$

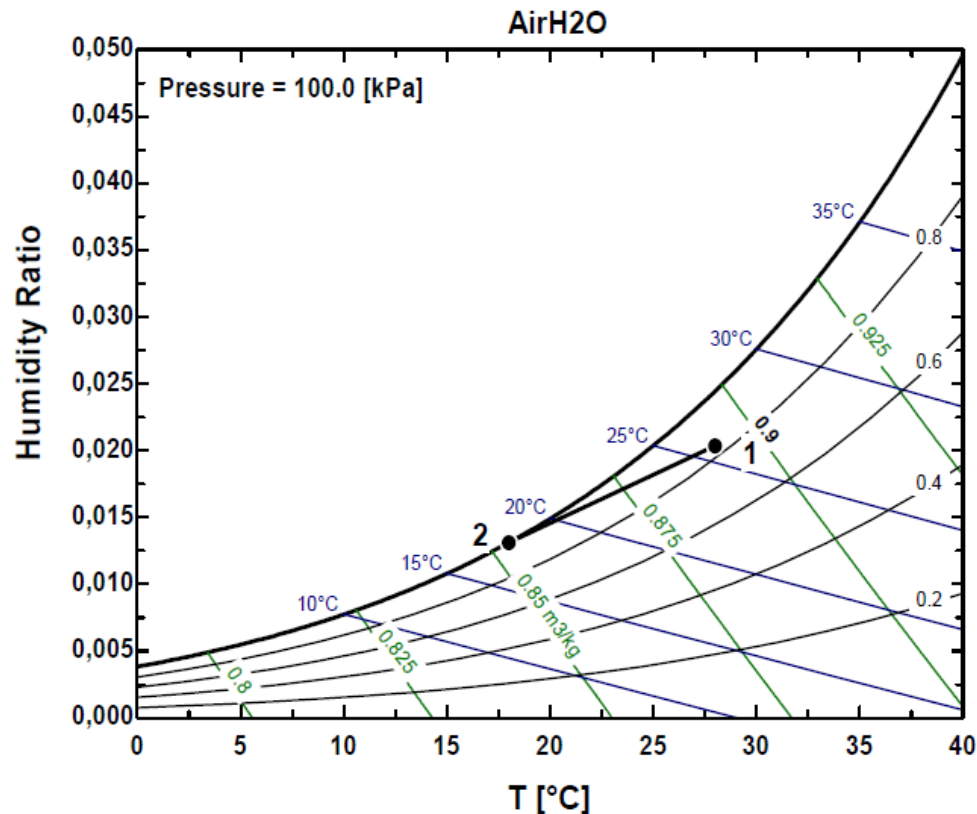
$$\dot{Q}_{\text{out}} = 1055 \text{ kJ/min} = 17.59 \text{ kW}$$

Noting that the rate of heat lost from the air is received by the cooling water, the mass flow rate of the cooling water is determined from

$$\dot{Q}_{\text{in}} = \dot{m}_{cw} c_p \Delta T_{cw} \longrightarrow \dot{m}_{cw} = \frac{\dot{Q}_{\text{in}}}{c_p \Delta T_{cw}} = \frac{1055 \text{ kJ/min}}{(4.18 \text{ kJ/kg} \cdot ^\circ\text{C})(10^\circ\text{C})} = 25.24 \text{ kg/min}$$

where we used the specific heat of water value at room temperature. Assuming a density of 1000 kg/m^3 for water, the volume flow rate is determined to be

$$\dot{V}_{cw} = \frac{\dot{m}_{cw}}{\rho_{cw}} = \frac{25.24 \text{ kg/min}}{1000 \text{ kg/m}^3} = 0.0252 \text{ m}^3/\text{min}$$



14-88 An automobile air conditioner uses refrigerant-134a as the cooling fluid. The evaporator operates at 275 kPa gauge and the condenser operates at 1.7 MPa gage. The compressor requires a power input of 6 kW and has an isentropic efficiency of 85 percent. Atmospheric air at 22°C and 50 percent relative humidity enters the evaporator and leaves at 8°C and 90 percent relative humidity. Determine the volume flow rate of the atmospheric air entering the evaporator of the air conditioner, in m³/min.

14-89 Air from a workspace enters an air conditioner unit at 30°C dry bulb and 25°C wet bulb. The air leaves the air conditioner and returns to the space at 25°C dry-bulb and 6.5°C dew-point temperature. If there is any, the condensate leaves the air conditioner at the temperature of the air leaving the cooling coils. The volume flow rate of the air returned to the workspace is 1000 m³/min. Atmospheric pressure is 98 kPa. Determine the heat transfer rate from the air, in kW, and the mass flow rate of condensate water, if any, in kg/h.

Evaporative Cooling

14-93 Air enters an evaporative cooler at 1 atm, 36°C, and 20 percent relative humidity at a rate of 4 m³/min, and it leaves with a relative humidity of 90 percent. Determine (a) the exit temperature of the air and (b) the required rate of water supply to the evaporative cooler.

Analysis (a) From the psychrometric chart (Fig. A-31) at 36°C and 20% relative humidity we read

$$T_{wb1} = 19.5^\circ\text{C}$$

$$\omega_1 = 0.0074 \text{ kg H}_2\text{O/kg dry air}$$

$$\nu_1 = 0.887 \text{ m}^3/\text{kg dry air}$$

Assuming the liquid water is supplied at a temperature not much different than the exit temperature of the air stream, the evaporative cooling process follows a line of constant wet-bulb temperature. That is,

$$T_{wb2} \cong T_{wb1} = 19.5^\circ\text{C}$$

At this wet-bulb temperature and 90% relative humidity we read

$$T_2 = 20.5^\circ\text{C}$$

$$\omega_2 = 0.0137 \text{ kg H}_2\text{O / kg dry air}$$

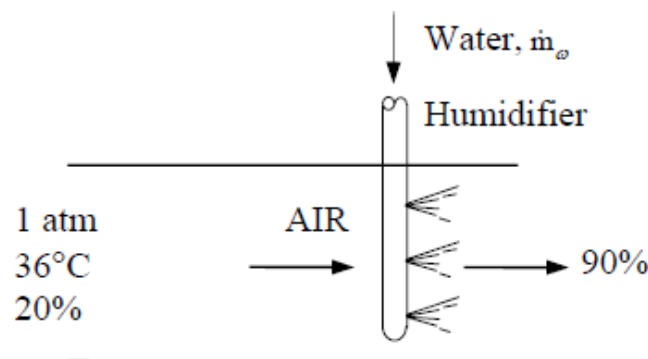
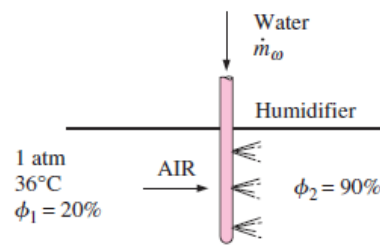
Thus air will be cooled to 20.5°C in this evaporative cooler.

(b) The mass flow rate of dry air is

$$\dot{m}_a = \frac{\dot{V}_1}{\nu_1} = \frac{4 \text{ m}^3/\text{min}}{0.887 \text{ m}^3/\text{kg dry air}} = 4.51 \text{ kg/min}$$

Then the required rate of water supply to the evaporative cooler is determined from

$$\begin{aligned} \dot{m}_{\text{supply}} &= \dot{m}_{w2} - \dot{m}_{w1} = \dot{m}_a (\omega_2 - \omega_1) \\ &= (4.51 \text{ kg/min})(0.0137 - 0.0074) = \mathbf{0.028 \text{ kg/min}} \end{aligned}$$

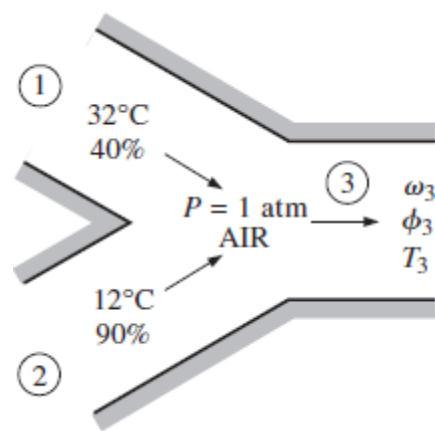


14-97 Air enters an evaporative cooler at 1 atm, 32°C, and 30 percent relative humidity at a rate of 5 m³/min and leaves at 22°C. Determine (a) the final relative humidity and (b) the amount of water added to air.

14-99 Air at 1 atm, 15°C, and 60 percent relative humidity is first heated to 30°C in a heating section and then passed through an evaporative cooler where its temperature drops to 25°C. Determine (a) the exit relative humidity and (b) the amount of water added to air, in kg H₂O/kg dry air.

Adiabatic Mixing of Airstreams

14-102 Two airstreams are mixed steadily and adiabatically. The first stream enters at 32°C and 40 percent relative humidity at a rate of 20 m³/min, while the second stream enters at 12°C and 90 percent relative humidity at a rate of 25 m³/min. Assuming that the mixing process occurs at a pressure of 1 atm, determine the specific humidity, the relative humidity, the dry-bulb temperature, and the volume flow rate of the mixture. *Answers: 0.0096 kg H₂O/kg dry air, 63.4 percent, 20.6°C, 45.0 m³/min*



Assumptions 1 Steady operating conditions exist 2 Dry air and water vapor are ideal gases. 3 The kinetic and potential energy changes are negligible. 4 The mixing section is adiabatic.

Properties Properties of each inlet stream are determined from the psychrometric chart (Fig. A-31) to be

$$h_1 = 62.7 \text{ kJ/kg dry air}$$

$$\omega_1 = 0.0119 \text{ kg H}_2\text{O/kg dry air}$$

$$\nu_1 = 0.882 \text{ m}^3/\text{kg dry air}$$

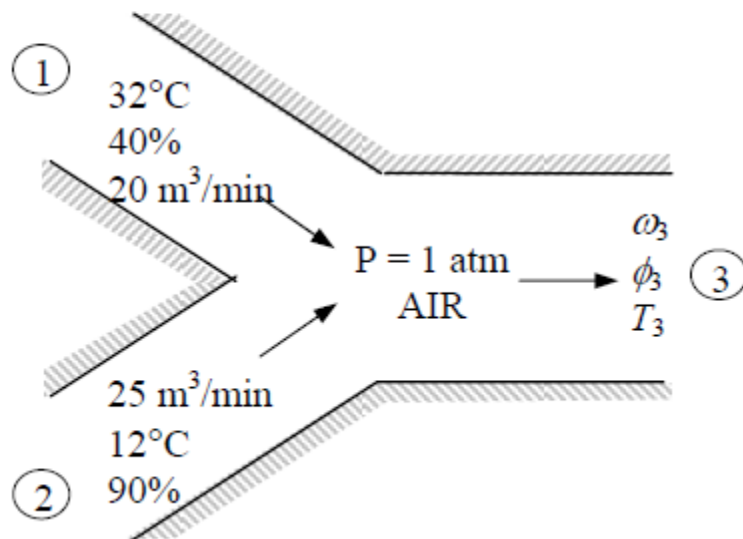
and

$$h_2 = 31.9 \text{ kJ/kg dry air}$$

$$\omega_2 = 0.0079 \text{ kg H}_2\text{O/kg dry air}$$

$$\nu_2 = 0.819 \text{ m}^3/\text{kg dry air}$$

Analysis The mass flow rate of dry air in each stream is



$$\dot{m}_{a1} = \frac{\dot{V}_1}{\nu_1} = \frac{20 \text{ m}^3 / \text{min}}{0.882 \text{ m}^3 / \text{kg dry air}} = 22.7 \text{ kg/min}$$

$$\dot{m}_{a2} = \frac{\dot{V}_2}{\nu_2} = \frac{25 \text{ m}^3 / \text{min}}{0.819 \text{ m}^3 / \text{kg dry air}} = 30.5 \text{ kg/min}$$

From the conservation of mass,

$$\dot{m}_{a3} = \dot{m}_{a1} + \dot{m}_{a2} = (22.7 + 30.5) \text{ kg / min} = 53.2 \text{ kg / min}$$

The specific humidity and the enthalpy of the mixture can be determined from Eqs. 14-24, which are obtained by combining the conservation of mass and energy equations for the adiabatic mixing of two streams:

$$\frac{\dot{m}_{a1}}{\dot{m}_{a2}} = \frac{\omega_2 - \omega_3}{\omega_3 - \omega_1} = \frac{h_2 - h_3}{h_3 - h_1}$$

$$\frac{22.7}{30.5} = \frac{0.0079 - \omega_3}{\omega_3 - 0.0119} = \frac{31.9 - h_3}{h_3 - 62.7}$$

which yields,

$$\omega_3 = 0.0096 \text{ kg H}_2\text{O} / \text{kg dry air}$$

$$h_3 = 45.0 \text{ kJ / kg dry air}$$

These two properties fix the state of the mixture. Other properties of the mixture are determined from the psychrometric chart:

$$T_3 = 20.6^\circ\text{C}$$

$$\phi_3 = 63.4\%$$

$$\nu_3 = 0.845 \text{ m}^3 / \text{kg dry air}$$

Finally, the volume flow rate of the mixture is determined from

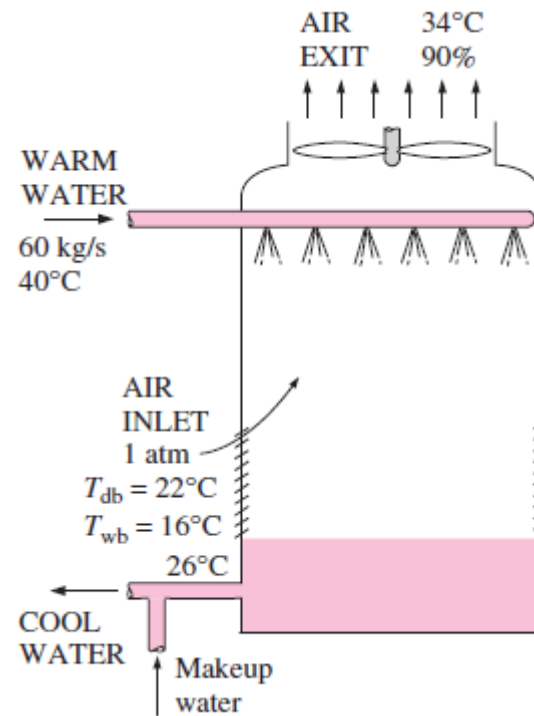
$$\dot{V}_3 = \dot{m}_{a3} \nu_3 = (53.2 \text{ kg/min})(0.845 \text{ m}^3 / \text{kg}) = 45.0 \text{ m}^3 / \text{min}$$

14–103 Repeat Prob. 14–102 for a total mixing-chamber pressure of 90 kPa.

14–106 A stream of warm air with a dry-bulb temperature of 40°C and a wet-bulb temperature of 32°C is mixed adiabatically with a stream of saturated cool air at 18°C . The dry air mass flow rates of the warm and cool airstreams are 8 and 6 kg/s, respectively. Assuming a total pressure of 1 atm, determine (a) the temperature, (b) the specific humidity, and (c) the relative humidity of the mixture.

Wet Cooling Towers

14-112 A wet cooling tower is to cool 60 kg/s of water from 40 to 26°C. Atmospheric air enters the tower at 1 atm with dry- and wet-bulb temperatures of 22 and 16°C, respectively, and leaves at 34°C with a relative humidity of 90 percent. Using the psychrometric chart, determine (a) the volume flow rate of air into the cooling tower and (b) the mass flow rate of the required makeup water. *Answers:* (a) 44.9 m³/s, (b) 1.16 kg/s



Analysis (a) The mass flow rate of dry air through the tower remains constant ($\dot{m}_{a1} = \dot{m}_{a2} = \dot{m}_a$), but the mass flow rate of liquid water decreases by an amount equal to the amount of water that vaporizes in the tower during the cooling process. The water lost through evaporation must be made up later in the cycle to maintain steady operation. Applying the mass and energy balances yields

Dry Air Mass Balance:

$$\sum \dot{m}_{a,i} = \sum \dot{m}_{a,e} \longrightarrow \dot{m}_{a1} = \dot{m}_{a2} = \dot{m}_a$$

Water Mass Balance:

$$\sum \dot{m}_{w,i} = \sum \dot{m}_{w,e} \rightarrow \dot{m}_3 + \dot{m}_{a1}\omega_1 = \dot{m}_4 + \dot{m}_{a2}\omega_2$$

$$\dot{m}_3 - \dot{m}_4 = \dot{m}_a(\omega_2 - \omega_1) = \dot{m}_{\text{makeup}}$$

Energy Balance:

$$\dot{E}_{\text{in}} - \dot{E}_{\text{out}} = \Delta \dot{E}_{\text{system}} \approx 0 \text{ (steady)} = 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\sum \dot{m}_i h_i = \sum \dot{m}_e h_e \quad (\text{since } \dot{Q} = \dot{W} = 0)$$

$$0 = \sum \dot{m}_e h_e - \sum \dot{m}_i h_i$$

$$0 = \dot{m}_{a2}h_2 + \dot{m}_4h_4 - \dot{m}_{a1}h_1 - \dot{m}_3h_3$$

$$0 = \dot{m}_a(h_2 - h_1) + (\dot{m}_3 - \dot{m}_{\text{makeup}})h_4 - \dot{m}_3h_3$$

Solving for $\dot{m}_a$,

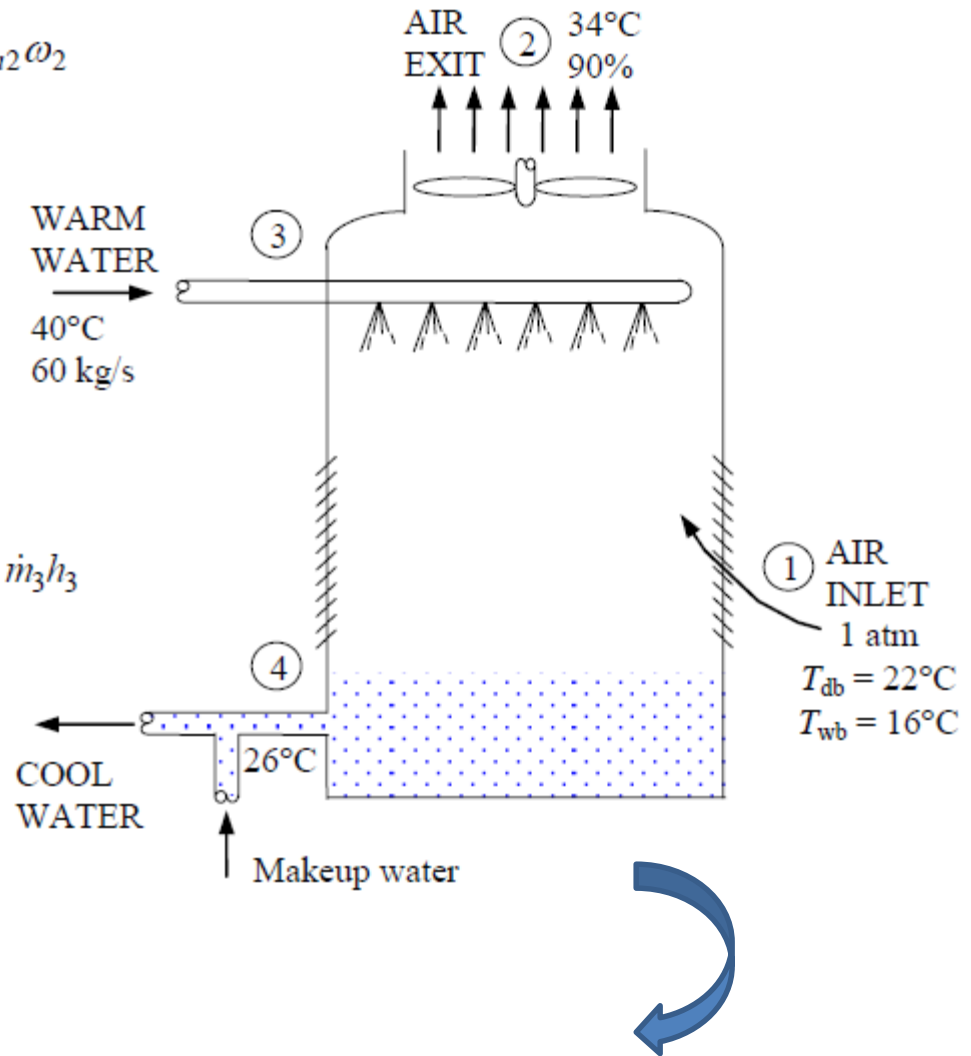
$$\dot{m}_a = \frac{\dot{m}_3(h_3 - h_4)}{(h_2 - h_1) - (\omega_2 - \omega_1)h_4}$$

From the psychrometric chart (Fig. A-31),

$$h_1 = 44.7 \text{ kJ/kg dry air}$$

$$\omega_1 = 0.0089 \text{ kg H}_2\text{O/kg dry air}$$

$$\nu_1 = 0.849 \text{ m}^3/\text{kg dry air}$$



and

$$h_2 = 113.5 \text{ kJ / kg dry air}$$

$$\omega_2 = 0.0309 \text{ kg H}_2\text{O / kg dry air}$$

From Table A-4,

$$h_3 \cong h_f @ 40^\circ\text{C} = 167.53 \text{ kJ/kg H}_2\text{O}$$

$$h_4 \cong h_f @ 26^\circ\text{C} = 109.01 \text{ kJ/kg H}_2\text{O}$$

Substituting,

$$\dot{m}_a = \frac{(60 \text{ kg/s})(167.53 - 109.01) \text{ kJ/kg}}{(113.5 - 44.7) \text{ kJ/kg} - (0.0309 - 0.0089)(109.01) \text{ kJ/kg}} = 52.9 \text{ kg/s}$$

Then the volume flow rate of air into the cooling tower becomes

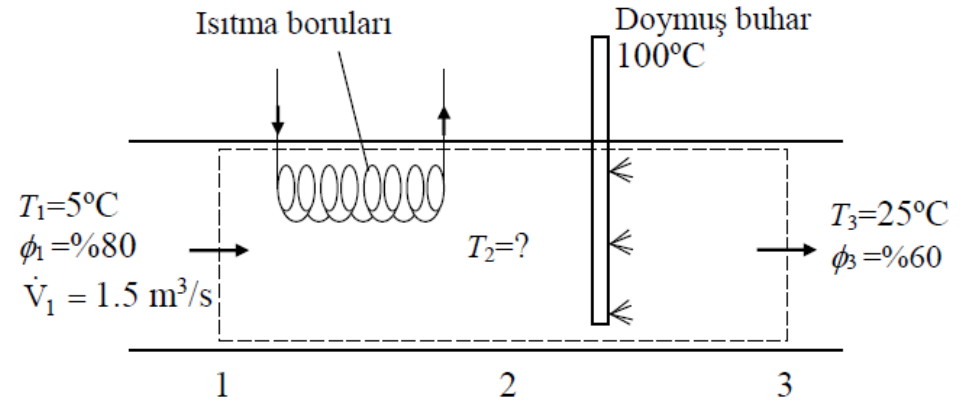
$$\dot{V}_1 = \dot{m}_a \nu_1 = (52.9 \text{ kg/s})(0.849 \text{ m}^3 / \text{kg}) = \mathbf{44.9 \text{ m}^3/\text{s}}$$

(b) The mass flow rate of the required makeup water is determined from

$$\dot{m}_{\text{makeup}} = \dot{m}_a (\omega_2 - \omega_1) = (52.9 \text{ kg / s})(0.0309 - 0.0089) = \mathbf{1.16 \text{ kg / s}}$$

Bir iklimlendirme sistemi 1 atm toplam basınçta çalışmaktadır. Sistem, şekilde görüldüğü gibi sürekli akışlı olup, bir ısıtma bölümü, bir de nemlendirme bölümünden oluşmaktadır. Nemli hava sisteme, 5°C sıcaklıkta ve %80 bağıl nemde, $1.5 \text{ m}^3/\text{s}$ debi ile girmekte, 25°C sıcaklıkta, %60 bağıl nemde çıkmaktadır. Nemlendirme 100°C sıcaklıkta doymuş su buharı kullanarak yapılmaktadır. **Psikrometrik diyagramı kullanmadan,**

- Sistemde akan kuru havanın kütle debisini, (kg/s)
- Isıtıcıdan geçtikten sonra nemli havanın sıcaklığını,
- Nemli havaya olan toplam ısı geçişini, (kW)
- Nemli havaya eklenen su buharının kütle debisini, (kg/s) hesaplayınız.
- Nemli havanın hal değişimini *psikrometrik diyagram* üzerinde gösteriniz.
- Nemli havanın giriş ve çıkıştaki entalpilerini (kJ/kg_a), *psikrometrik diyagramı* kullanarak bulunuz.



[İPUCU: Önce nemlendirme bölümünü ele alın, adyabatik olduğunu varsayın ve $h_{v2} = 2540 \text{ kJ/kg}$, $c_{pa} = 1.005 \text{ kJ/kg-K}$ alın.]

$$\begin{aligned}
 \text{(a)} \quad & \left. \begin{aligned} T_1 &= 5^\circ\text{C} \\ \phi_1 &= \%80 \\ \dot{V}_1 &= 1.5 \text{ m}^3/\text{s} \end{aligned} \right\} \begin{aligned} P_{g1} &= P_{\text{doyma},5\text{C}} = 0.8725 \text{ kPa}, \quad h_{v1} \cong h_{g,5\text{C}} = 2510.1 \text{ kJ/kg} \quad (\text{Tablo A-4}) \\ P_{v1} &= \phi_1 P_{\text{doyma},5\text{C}} = 0.80 \times 0.8725 = 0.698 \text{ kPa} \\ P_{a1} &= P_1 - P_{v1} = 101.325 - 0.698 = 100.63 \text{ kPa} \\ \omega_1 &= 0.622 \frac{P_{v1}}{P_{a1}} = 0.622 \times \frac{0.698}{100.63} = 0.0043 \text{ kg}_v/\text{kg}_a \\ h_1 &= h_{a1} + \omega_1 h_{v1} \cong h_{a1} + \omega_1 h_{g1} = 1.005 \times 5 + 0.0043 \times 2510.1 = 15.82 \text{ kJ/kg}_a \\ v_{a1} &= \frac{R_a T_1}{P_{a1}} = \frac{0.287 \times 278}{100.63} = 0.7929 \text{ m}^3/\text{kg}_a \quad (\rho_{a1} = 1/v_{a1} = 1.2612 \text{ kg}_a/\text{m}^3) \\ \dot{m}_a &= \frac{\dot{V}_1}{v_{a1}} = \frac{1.5}{0.7929} = 1.89 \text{ kg/s} \quad (\dot{m}_{a1} = \dot{m}_{a2} = \dot{m}_{a3} = \dot{m}_a = 1.89 \text{ kg/s}) \end{aligned}
 \end{aligned}$$

$$\begin{aligned}
 & P_{g3} = P_{\text{doyma},25\text{C}} = 3.1698 \text{ kPa}, \quad h_{v3} \cong h_{g,25\text{C}} = 2546.5 \text{ kJ/kg} \quad (\text{Tablo A-4}) \\
 \text{(b)} \quad & \left. \begin{aligned} T_3 &= 25^\circ\text{C} \\ \phi_3 &= \%60 \end{aligned} \right\} \begin{aligned} P_{v3} &= \phi_3 P_{\text{doyma},25\text{C}} = 0.60 \times 3.1698 = 1.9019 \text{ kPa} \\ P_{a3} &= P_3 - P_{v3} = 101.325 - 1.9019 = 99.42 \text{ kPa} \\ \omega_3 &= 0.622 \frac{P_{v3}}{P_{a3}} = 0.622 \times \frac{1.9019}{99.42} = 0.0119 \text{ kg}_v/\text{kg}_a \\ h_3 &= h_{a3} + \omega_3 h_{v3} \cong h_{a3} + \omega_3 h_{g3} = 1.005 \times 25 + 0.0119 \times 2546.5 = 55.43 \text{ kJ/kg}_a \end{aligned}
 \end{aligned}$$

Isıtıcıda $\omega = \text{sabit}$ olduğundan : $\omega_2 = \omega_1 = 0.0043 \text{ kg}_v/\text{kg}_a$

Isıtıcı çıkışında nemli havanın entalpisi (h_2) : $h_2 = h_{a2} + \omega_2 h_{v2} = 1.005 T_2 + 0.0043 \times 2540 = 1.005 T_2 + 10.92$

$T_w = 100^\circ\text{C}$ sıcaklıkta doymuş su buharının entalpisi : $h_w = h_{g,100\text{C}} = 2675.6 \text{ kJ/kg}$

Açık Sistem : Nemlendirici (SASA)

Kütlenin korunumu Kuru hava: $\dot{m}_{a2} = \dot{m}_{a3} = \dot{m}_a = 1.89 \text{ kg/s}$

Su :

$$\dot{m}_{v3} = \dot{m}_{v2} + \dot{m}_w$$

$\Rightarrow$

$$\dot{m}_w = \dot{m}_{v3} - \dot{m}_{v2} = \dot{m}_{a3} \omega_3 - \dot{m}_{a2} \omega_2 = \dot{m}_a (\omega_3 - \omega_2)$$

$$1. \text{ Yasa } (\Delta ke \cong 0, \Delta pe \cong 0, \dot{Q} \cong 0, \dot{W} = 0)$$

$$\dot{Q}^{\text{so}} - \dot{W}^{\text{so}} = \dot{m}_{a3}h_3 - \dot{m}_{a2}h_2 - \dot{m}_w h_w \Rightarrow 0 = \dot{m}_{a3}h_3 - \dot{m}_{a2}h_2 - \dot{m}_w h_w$$

$$0 = \dot{m}_a (h_3 - h_2) - \dot{m}_a (\omega_3 - \omega_2) h_w \Rightarrow h_2 = h_3 - (\omega_3 - \omega_2) h_w$$

$$h_2 = 55.43 - \underbrace{(0.0119 - 0.0043) \times 2675.6}_{0.0076} = 35.1 \text{ kJ / kg}_a$$

$$\text{Böylece } h_2 = 1.005 T_2 + 10.92 = 35.1 \text{ kJ / kg}_a \Rightarrow T_2 = \frac{35.1 - 10.92}{1.005} = \frac{24.18}{1.005} \cong 24.1^\circ \text{C}$$

$$[P_{g2} = P_{\text{doyma}, 24.1^\circ \text{C}} \cong 3 \text{ kPa}, \quad \phi_2 = \frac{\omega_2 P_2}{(0.622 + \omega_2) P_{g2}} = \frac{0.0043 \times 101.325}{(0.622 + 0.0043) \times 3} \cong 0.23 \quad (\%23) \quad \text{SORULMUYOR}$$

(c) Açık Sistem : Tüm iklimlendirme Cihazı (SASA)

Kütlenin korunumu Kuru hava: $\dot{m}_{a3} = \dot{m}_{a1} = \dot{m}_a = 1.89 \text{ kg / s}$

$$\text{Su} : \dot{m}_{v3} = \dot{m}_{v1} + \dot{m}_w$$

$$\dot{m}_w = \dot{m}_{v3} - \dot{m}_{v1} = \dot{m}_{a3}\omega_3 - \dot{m}_{a1}\omega_1 = \dot{m}_a (\omega_3 - \omega_1)$$

$$1. \text{ Yasa } (\Delta ke \cong 0, \Delta pe \cong 0, \dot{W} = 0)$$

$$\dot{Q} - \dot{W}^{\text{so}} = \dot{m}_{a3}h_3 - \dot{m}_{a1}h_1 - \dot{m}_w h_w \Rightarrow \dot{Q} = \dot{m}_{a3}h_3 - \dot{m}_{a2}h_2 - \dot{m}_w h_w$$

$$\dot{Q} = \dot{m}_a (h_3 - h_1) - \dot{m}_a (\omega_3 - \omega_1) h_w \Rightarrow \dot{Q} = \dot{m}_a [(h_3 - h_1) - (\omega_3 - \omega_1) h_w]$$

$$\begin{aligned} \dot{Q} &= 1.89 \times [(55.43 - 15.82) - (0.0119 - 0.0043) \times 2675.6] \\ &= 1.89 \times [39.61 - 20.33] \\ &= 1.89 \times 19.28 \\ &= 36.5 \text{ kW} \end{aligned}$$

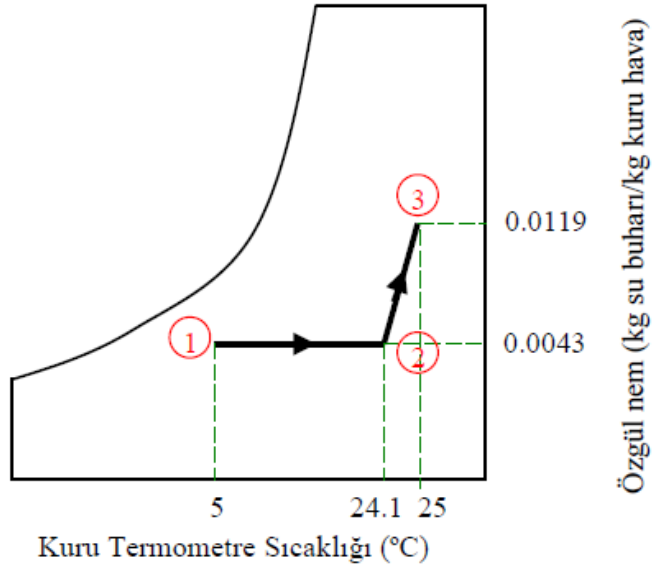
veya havaya olan ısı geçişi sadece ısıtma bölümünde olduğundan:

$$\text{Açık Sistem : Isıtma Bölümü (SASA)} \quad \dot{Q} = \dot{m}_a (h_2 - h_1) = 1.89 \times (35.1 - 15.82) = 36.5 \text{ kW}$$

(d) Nemli havaya eklenen su buharının kütle debisi :

$$\dot{m}_w = \dot{m}_{v3} - \dot{m}_{v2} = \dot{m}_a (\omega_3 - \omega_2)$$
$$\dot{m}_w = 1.89 \times \underbrace{(0.0119 - 0.0043)}_{0.0076} = 0.0144 \text{ kg / s}$$

(e)



(f) Psikrometrik diyagramdan $h_1 \cong 16 \text{ kJ / kg}_a$, $h_3 \cong 56 \text{ kJ / kg}_a$

Not: kg_v/kg_a : $\text{kg su buharı/kg kuru hava}$ ve kJ/kg_a : kJ/kg kuru hava anlamındadır.

Chapter 14

End of the Problems